

Project

Efficient Architecture for Direct Digital Frequency Synthesizers Using Phase Accumulator and LUT Optimization

❖ Block Diagram

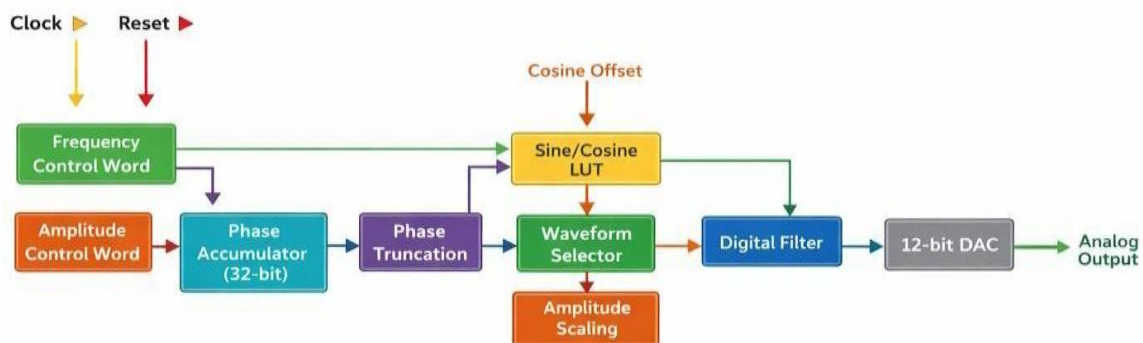


Fig 1: Block Diagram of DDFS

❖ Code:



❖ Figure:

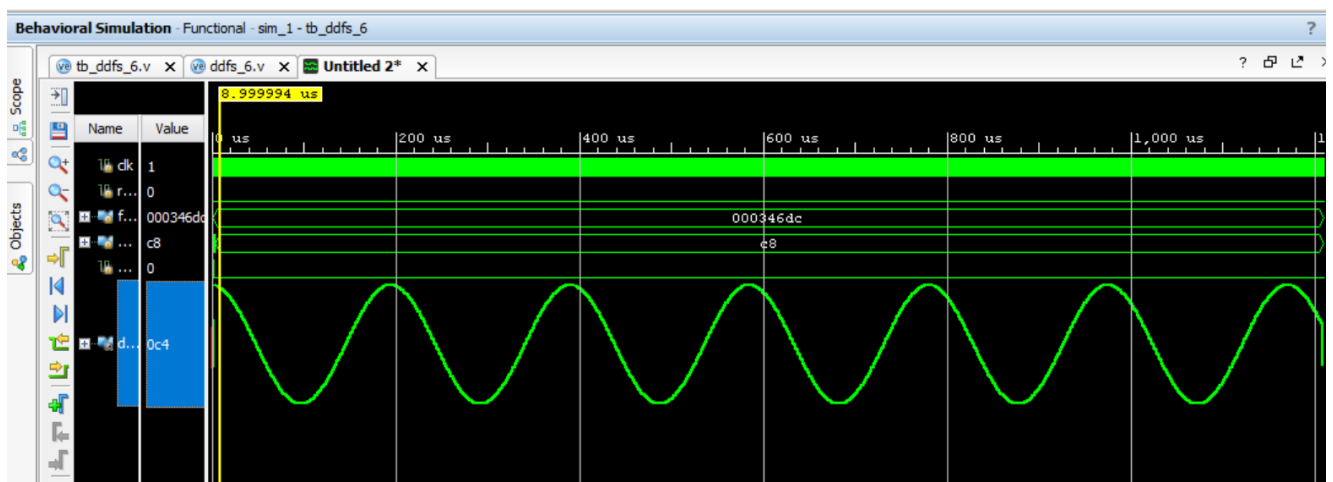


Fig 2: Simulation Waveform of DDFS

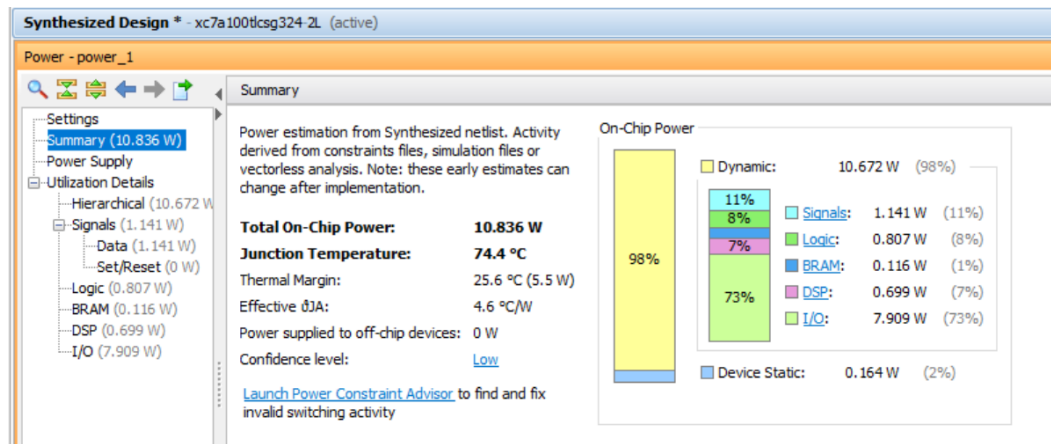


Fig 3: Power consumption of DDFS

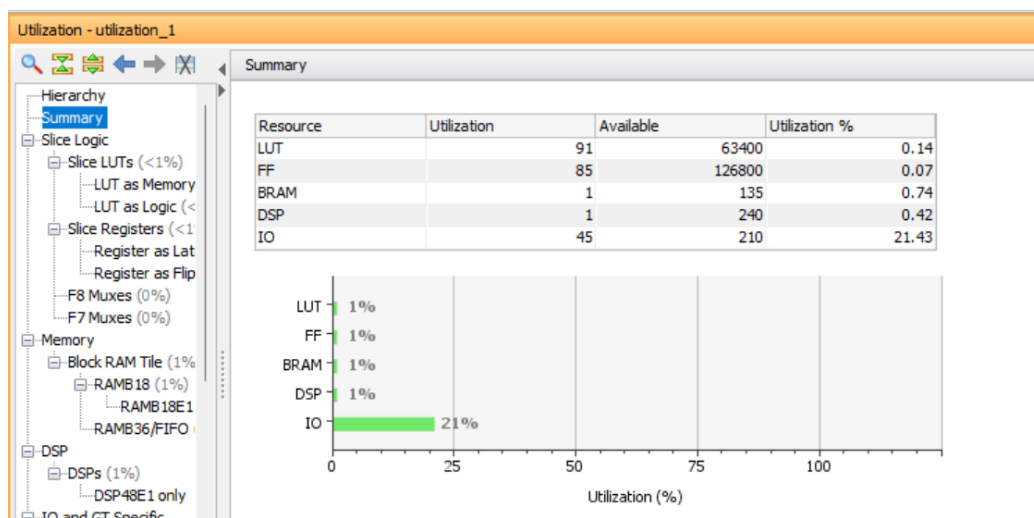


Fig 4: Area Utilization of DDFS

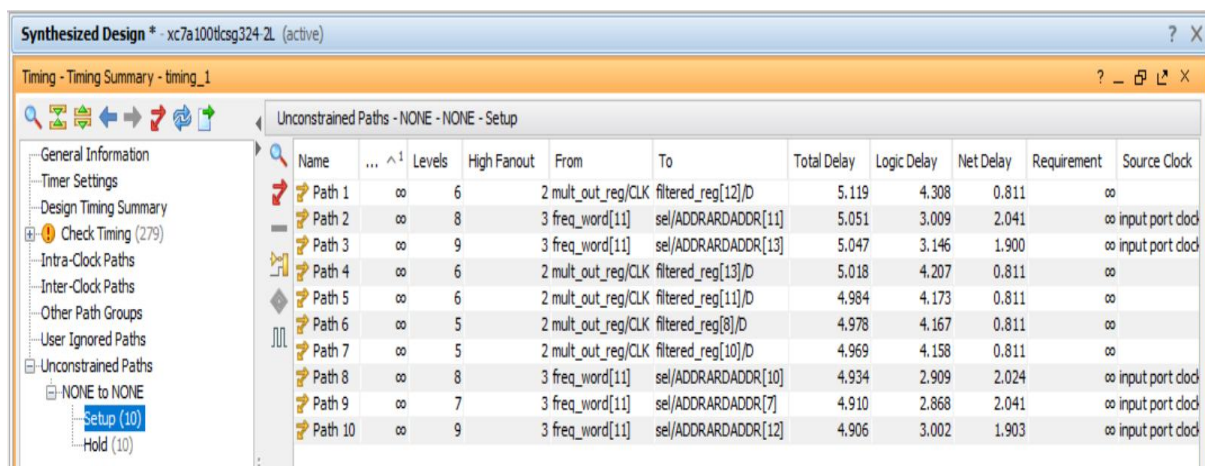


Fig 5: Delay of DDFS

❖ Formulas:

1. $\text{phase} \leq \text{phase} + 0.001 * \text{freq_word}$
2. $\text{freq_word} = 32'd214748; // 5\text{kHz}$
3. Clock frequency(f_{clk})= 10ns=100MHz
4. $N=32$

DDFS frequency formula:

$$F_{\text{out}} = (\text{freq_word} \times f_{\text{clk}}) / 2^N$$

Inverse:

$$\text{freq_word} = (f_{\text{out}} \times 2^N) / f_{\text{clk}}$$

given

$$\text{freq_word} = 32'd214748$$

$$f_{\text{clk}} = 100000000 // 100 \text{ MHz}$$

$$N = 32$$

$$2^N = 2^{32} = 4294967296$$

$$\text{Numerator} = 214748 \times 100000000 = 21400000000$$

$$f_{\text{out}} = 21400000000 / 4294967296 \approx 4980 \text{ Hz}$$

$$f_{\text{out}} = 5\text{KHz}$$